

Lab 3 Solution

```
library(tidyverse)

ncbikecrash <- read_csv("https://karamccor.github.io/b6002/labs/data/bikecrash.csv")
```

Exercise 1

(a)

```
ncbikecrash |>
  filter(development == "Residential") |>
  filter(driver_age_group == "0-19")
```

There were 223 crashes in residential areas where the driver age group was under 0-19.

(b)

```
ncbikecrash |>
  filter(development == "Residential") |>
  count(driver_age_group)
```

```
# A tibble: 9 x 2
  driver_age_group      n
  <chr>             <int>
1 0-19               223
2 20-24             364
3 25-29             290
4 30-39             508
5 40-49             441
6 50-59             372
```

7 60-69	269
8 70+	243
9 <NA>	457

Exercise 2

```
ncbikecrash |>
  count(driver_est_speed) |>
  arrange(n)
```

```
# A tibble: 17 x 2
  driver_est_speed      n
  <chr>           <int>
1 81-85 mph         1
2 >100 mph          2
3 66-70 mph         4
4 61-65 mph        10
5 56-60 mph        29
6 46-50 mph       142
7 36-40 mph       261
8 51-55 mph       338
9 26-30 mph       367
10 21-25 mph      525
11 16-20 mph      544
12 11-15 mph      552
13 <NA>           580
14 41-45 mph      670
15 6-10 mph       840
16 31-35 mph      846
17 0-5 mph      1756
```

Exercise 3

```
ncbikecrash |>
  filter(county == "Durham" | county == "Orange" | county == "Wake") |>
  group_by(county) |>
  summarize(mean_driver_age = mean(driver_age, na.rm = T))
```

```
# A tibble: 3 x 2
  county mean_driver_age
  <chr>      <dbl>
1 Durham      40.4
2 Orange      38.3
3 Wake        38.2
```

Exercise 4

```
ncbikecrash |>
  filter(county == "Durham") |>
  group_by(crash_month) |>
  summarize(mean_crash_hour = mean(crash_hour)) |>
  arrange(mean_crash_hour)
```

```
# A tibble: 12 x 2
  crash_month mean_crash_hour
  <chr>      <dbl>
1 January      13.5
2 August       13.9
3 September    14.0
4 October      14.2
5 November     14.4
6 December     14.6
7 May          14.9
8 March        15.2
9 July         15.3
10 April       15.3
11 June        16.2
12 February    16.8
```

Exercise 5

```
ncbikecrash_time <- ncbikecrash |>
  mutate(time_crash = case_when(
    crash_hour < 7 ~ "Early Morning",
    crash_hour >= 7 & crash_hour < 12 ~ "Morning",
    crash_hour >= 12 & crash_hour < 18 ~ "Afternoon",
```

```

    crash_hour >= 18 ~ "Evening"
  ))

ncbikecrash_time |>
  count(time_crash)

```

```

# A tibble: 4 x 2
  time_crash      n
  <chr>        <int>
1 Afternoon    3412
2 Early Morning 425
3 Evening     2270
4 Morning     1360

```

Exercise 6

```

ncbikecrash |>
  summarize(mean_driver_age = mean(driver_age, na.rm = T))

```

```

# A tibble: 1 x 1
  mean_driver_age
  <dbl>
1          39.0

```

```

ncbikecrash |>
  group_by(county) |>
  summarize(mean_driver_age = mean(driver_age, na.rm = T)) |>
  arrange(desc(mean_driver_age)) |>
  slice(1:5)

```

```

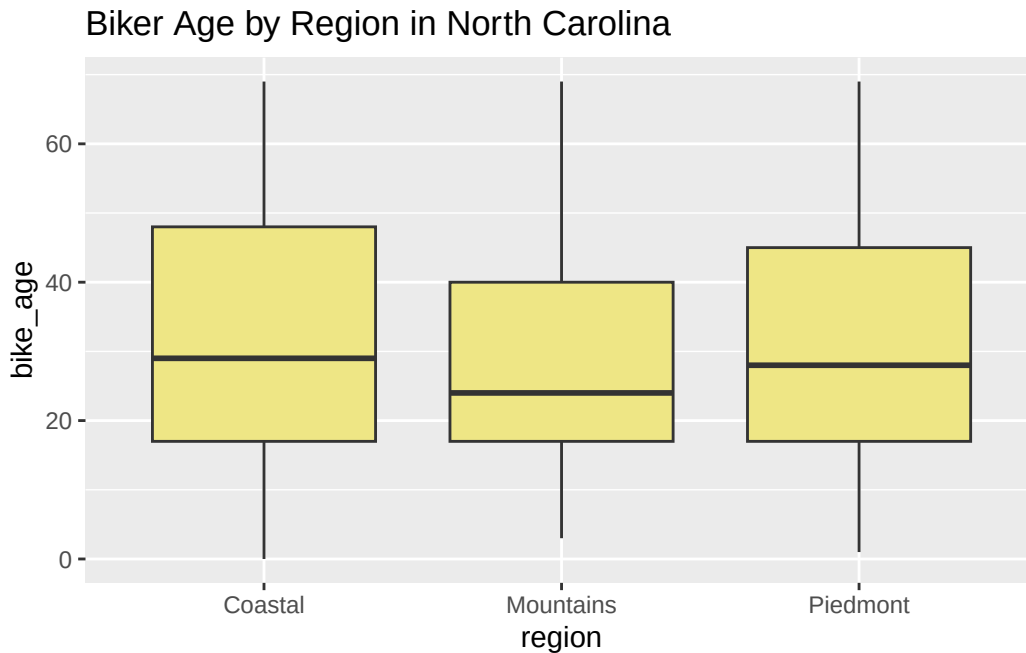
# A tibble: 5 x 2
  county mean_driver_age
  <chr>      <dbl>
1 Camden      57.8
2 Stokes      55.2
3 Mitchell     54
4 Anson       49.6
5 Polk        47

```

Exercise 7

(a)

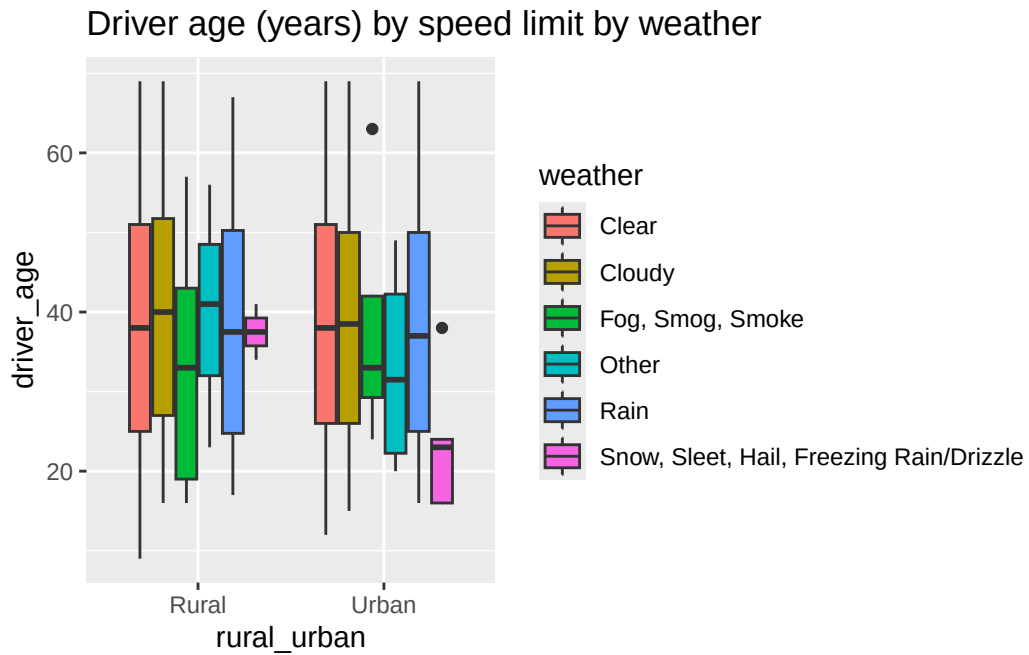
```
ggplot(ncbikecrash, mapping = aes(x = region, y = bike_age)) +  
  geom_boxplot(fill = "khaki2") +  
  labs(title = "Biker Age by Region in North Carolina",  
        xlab = "Urban vs. Rural area",  
        ylab = "Driver age (years)")
```



The biker ages seem to be relatively similar across groups.

(b)

```
ggplot(ncbikecrash, mapping = aes(x = rural_urban, y = driver_age, fill = weather)) +  
  geom_boxplot() +  
  labs(title = "Driver age (years) by speed limit by weather",  
        xlab = "Rural vs. Urban area",  
        ylab = "Driver Age (years)")
```



For Urban areas with **Snow, Sleet, Hail, Freezing Rain/Drizzle**, drivers involved in the crashes tended to be younger, with a median around 23 years old, compared to Rural areas where the same group was older (median age around 36 years.)